

Mathematical Foundations of Political Methodology Lecture 5

Robert Gulotty

Department of Political Science
University of Chicago

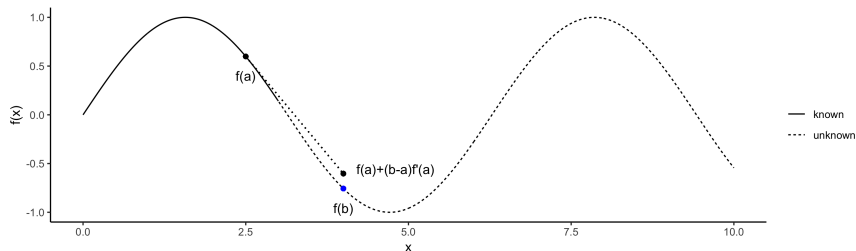
August 28th, 2017

- Last time we learned about the Mean Value Theorem.
- We will briefly return to that topic before turning to techniques for finding maxima and minima.
- How much can we infer about the value of a function at a point x given the curvature at a point x_0 ?

Linear Approximation.

- We want to know $f(b)$, we know $f(a)$, $f'(a)$
- With the slope and the distance between a and b we can calculate the position of $f(b)$:

$$f(b) \approx f(a) + (b - a)f'(a)$$



- Can we do better?

Preliminary Result: Second Mean Value Theorem (pg 189).

Theorem

If f is a twice-differentiable function and a and b are real numbers such that $a < b$, there exists d such that $a < d < b$ and

$$f(b) = f(a) + (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(d)$$

Applying the Second MVT

$$f(b) = f(a) + (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(d)$$

Set $b = a + h$, $d = a + th$ for some $0 < t < 1$:

$$f(a + h) = f(a) + (h)f'(a) + \frac{1}{2}(h)^2 f''(a + th)$$

If $|h|$ isn't too big:

$$f(a + h) \approx f(a) + (h)f'(a) + \frac{1}{2}(h)^2 f''(a)$$

Replace $h = x - a$,

$$\tilde{f}(x) = f(a) + (x - a)f'(a) + \frac{1}{2}(x - a)^2 f''(a)$$

$\tilde{f}(x)$ is called the quadratic approximation of $f(x)$

Moving to Taylor's Theorem

- The best approximation uses higher order derivatives.
- If a function has arbitrary many higher derivatives, it is called smooth.
- We use these higher derivatives to approximate a function to arbitrary precision.

Taylor's Theorem

If f is a smooth function and n is a positive integer:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{h^n}{(n)!}f^{(n)}(a+th)$$

The $(n-1)$ th order Taylor approximation sets the n th term to 0:

$$f(a+h) \approx f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + 0$$

Taylor's Series Expansion

In general, the Taylor's Approximations take advantage of Taylor's Series:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \frac{h^3}{3!}f'''(a) + \dots$$

is a convergent series.

- The Taylor Expansion plays a starring role in the Central Limit Theorem, the main result of the second half of this class.
- It is also used to characterize the exponential function e .

$$e^x = f(x) = f'(x) = f''(x) \dots$$

Note that $e^0 = 1$. Set $a = 0$

$$e^{0+h} = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} \dots$$

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

Monotone Functions

Definition

Monotonic functions: functions that are either strictly increasing or strictly decreasing

- Strictly increasing: If $x_2 > x_1$ then $f(x_2) > f(x_1)$
- Strictly decreasing: If $x_2 > x_1$ then $f(x_2) < f(x_1)$
- If $f'(x) > 0$ for all x , then f is strictly increasing; however this is stronger than we need for a function to be strictly increasing (e.g. $f(x) = x^3$)
- If $f'(x) \geq 0$ for all x and $f'(x) = 0$ for only a finite number of x , then f is strictly increasing (with strictly decreasing proved similarly)

Inverse Functions

Definition

Let $f(x)$ be monotonic. Then for any y , $f(x) = y$ has at most one solution. If the solution is written $x = g(y)$, then g is the inverse function of f , or

$$f(g(y)) = y \text{ and } g(f(x)) = x$$

Example 1: Let $f(x) = x^3$; what is the inverse function of f ?

Example 2: Let $f(x) = x^{-1} + 5$; what is the inverse function of f ?

The inverse function is a reflection of the function about the 45° line; what's the inverse of the function $\{(1, 2), (2, 3), (3, 4)\}$?

The inverse function rule

If f is monotonic with inverse function g , and for a given x , f is differentiable and $f'(x) \neq 0$, then

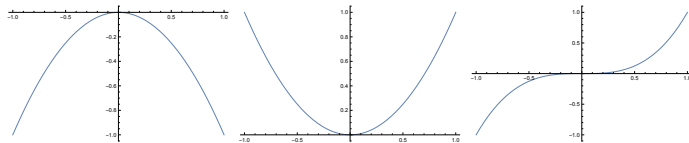
$$g'(y) = \frac{1}{f'(x)}$$

This gives us another technique for differentiating things; we could calculate f' directly, or we could calculate $\frac{1}{g'}$

Critical Points

If $f'(x) = 0$ then x is called a *critical point*

- Maximum
- Minimum
- Inflection point



Critical Points

One of the main things we use calculus for is to identify and classify critical points

- 1 Find $f'(x)$
- 2 Find the set of x such that $f'(x) = 0$
- 3 Intuitively, if $f'(x)$ changes from negative to positive around x , then x is a minimum; if it changes from positive to negative then x is a maximum; if it does neither then it's a point of inflection
- 4 Always remember that without more knowledge of f , all we know is that x is a *local* maximum or minimum (or an inflection point); we can't assume that it maximizes or minimizes our function globally

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

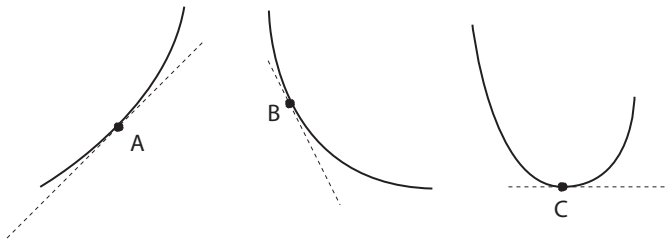
Second Derivative Test

Recall that if f' is itself differentiable, then f is *twice differentiable*, and $f'' = \frac{d^2y}{dx^2}$ is the *second derivative*

- Suppose that $f''(x)$ is positive at some point x
- This means that the *slope of the slope* of f is increasing at x
- This means that either:
 - If f is increasing then it's increasing at an increasingly faster rate at x
 - If f is decreasing then it's decreasing at an increasingly slower rate at x
 - If f is flat then it's increasing as x increases
- In all of these cases, f lies above its tangent

Second Derivative Test

- A If f is increasing then it's increasing at an increasingly faster rate at x
- B If f is decreasing then it's decreasing at an increasingly slower rate at x
- C If f is flat then it's increasing as x increases



Second Derivative Test

- For any critical point, x^* , calculate $f''(x^*)$
 - If $f''(x^*) > 0$ then x^* is a local minimum
 - If $f''(x^*) < 0$ then x^* is a local maximum
 - If $f''(x^*) = 0$ then we don't know what x^* is; we have to examine our function around x^*
- $f'(x) = 0$ is our first-order condition; conditions on f'' are our second-order conditions

Find and classify the critical points of $x^3 - 3x$, and then use this information to plot out the function

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

Global Maxima

If $f(x^*) \geq f(x)$ for all $x \in \mathbb{R}$ then x^* is a global maximum

- Global maxima survive “strictly increasing functions”.
 - Suppose that $g(x) = H(f(x))$ where H is a *strictly increasing function*. Then if f has a global maximum at x^* , $g(x)$ also has a global maximum at x^*
 - One useful strictly increasing function is the natural log:
 - If x^* globally maximizes $\log(f(x))$ then x^* also maximizes $f(x)$
- If x^* is the global max of $f(x)$ then x^* is the global minimum of $-f(x)$

Existence of global maximum and minimum

Theorem

The extreme value theorem: any continuous function on a closed and bounded interval attains a global maximum and minimum

Boundary Conditions & Asymptotes

- Often functions have max and min at the boundaries or at asymptotes:

$y = \frac{1}{x}$ ($x > 0$) has a x and y asymptotes equal to the axes

- An important strategy is to trace out functions.

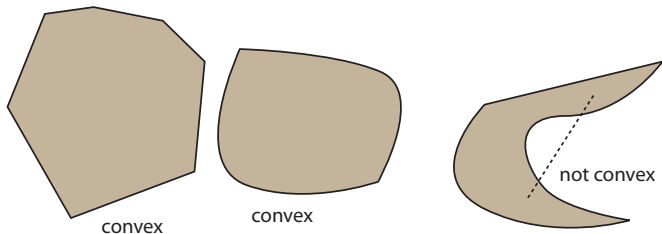
Trace out: $y = x - \frac{1}{x}$

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

Convexity and Concavity

Definition

A set is convex if the line segment connecting any two points in the set is entirely contained within the set

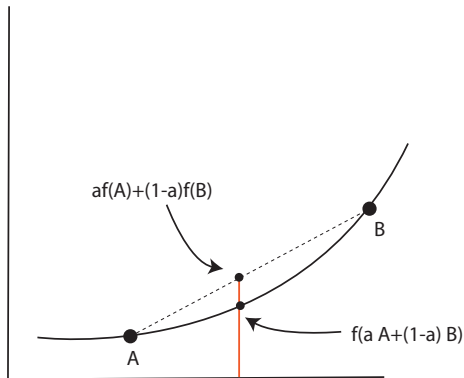


Convex Function

Definition

f is a convex function if and only if for all real numbers A, B and all $0 \leq \alpha \leq 1$,

$$\alpha f(A) + (1 - \alpha)f(B) \geq f(\alpha A + (1 - \alpha)B)$$



Properties of Convex Functions

Convex functions need not be differentiable (e.g. $y = |x|$).

- If f is twice differentiable, then f is convex if and only if $f'' \geq 0$ for all x
- If f is differentiable and convex then
 - There is either no global minimum, one global minimum, or an interval of global minima
 - There are no local minima that are not global minima
 - If $f'(x^*) = 0$ then x^* is a global minimum

Concave Function

Definition

f is concave if and only if $-f$ is convex: the set of all points on or below f is convex

- Concave functions are typical choices for utility functions.

Properties of Concave Function

- If f is twice differentiable then f is concave if and only if $f'' \leq 0$ for all x .
- If f is concave and $f'(x^*) = 0$ then x^* is a global maximum.
- To remember, **concave** looks like the entrance to a cave.

Find Maxima/Minima: A Summary

- Compute $f'(x)$ and find all x such that $f'(x) = 0$;
- Conduct second derivative test, compute $f''(x)$ for all x such that $f'(x) = 0$;
 - $f''(x) > 0$, then, it's a local minimum;
 - $f''(x) < 0$, then, it's a local maximum;
 - $f''(x) = 0$, then, undecided.
- Finally, check boundary points, and determine global maxima or minima.

Example 1: $f(x) = -x^2$, $x \in [-3, 3]$

1) Critical Value:

$$\begin{aligned}f'(x) &= -2x \\0 &= -2x^* \\x^* &= 0\end{aligned}$$

2) Second Derivative:

$$\begin{aligned}f'(x) &= -2x \\f''(x) &= -2\end{aligned}$$

$f''(x) < 0$, local maximum

Example 1: $f(x) = -x^2$, $x \in [-3, 3]$

3) Check end points

$$\begin{aligned} f(0) &= -0^2 = 0 \\ f(-3) &= -(-3)^2 = -9 \\ f(3) &= -(3)^2 = -9 \end{aligned}$$

Example 2: $f(x) = x^3$, $x \in [-3, 3]$

1) Critical Values:

$$\begin{aligned}f'(x) &= 3x^2 \\ 0 &= 3(x^*)^2 \\ x^* &= 0\end{aligned}$$

2) Second Derivative:

$$\begin{aligned}f''(x) &= 6x \\ f''(0) &= 0\end{aligned}$$

No information

Example 2: $f(x) = x^3$, $x \in [-3, 3]$

3) Check End Points:

$$f(0) = 0^3 = 0$$

$$f(-3) = -3^3 = -27$$

$$f(3) = 3^3 = 27$$

Neither maximum nor minimum, saddle point

Example 3: Spatial Model

A large literature in Congress supposes legislators and policies can be situated in **policy space**

Suppose legislator i and policies $x, i \in \mathbb{R}$.

Define legislator i 's utility as, $U : \mathbb{R} \rightarrow \mathbb{R}$,

$$U_i(x) = -(x - \mu)^2$$

$$U_i(x) = -x^2 + 2x\mu - \mu^2$$

What is i 's optimal policy over the range $x \in [\mu - 2, \mu + 2]$?

$$U'_i(x) = -2(x - \mu)$$

$$0 = -2x^* + 2\mu$$

$$x^* = \mu$$

Second Derivative Test

$$U''_i(x) = -2 < 0$$

We call μ legislator i 's **ideal point**

Example 3: Spatial Model

$$\begin{aligned}U_i(\mu) &= -(\mu - \mu)^2 = 0 \\U_i(\mu - 2) &= -(\mu - 2 - \mu)^2 = -4 \\U_i(\mu + 2) &= -(\mu + 2 - \mu)^2 = -4\end{aligned}$$

Maximize utility at μ